ARJUNA JEE 2027

Lecture -08 | Mathematics

BINOMIAL THEOREM



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Topics *to be covered*

A Application of Binomial Theorem

B Finding Remainder

C Integral & Fractional Part

D Question Practice



H.W. Discussion

Question [JEE Mains-2022]

73%

$\sum_{\substack{i,j=0 \\ i\neq j}}^{n} {}^{n}C_i \, {}^{n}C_j$ is equal to

$$\sum_{j=0}^{n}\sum_{\substack{i=0 \\ i\neq j}}^{n} {}^{n}C_i \, {}^{n}C_j = \sum_{0\le i<j\le n}\sum {}^{n}C_i \, {}^{n}C_j + \sum_{0\le j<i\le n}\sum {}^{n}C_i \, {}^{n}C_j$$

A $2^{2n} - {}^{2n}C_n$

B $2^{2n-1} - {}^{2n-1}C_{n-1}$

C $2^n - \frac{1}{2}{}^{2n}C_n$

D $2^{n-1} + {}^{2n-1}C_n$

$$\text{Total} = \sum_{j=0}^{n}\left(\sum_{i=0}^{n} {}^{n}C_i \, {}^{n}C_j\right) = \sum_{j=0}^{n}\left({}^{n}C_j \left(\sum_{i=0}^{n} {}^{n}C_i\right)\right) = 2^n \sum_{j=0}^{n} {}^{n}C_j = 2^{2n}$$

2^n

$$\text{Total} = \sum_{i=j} \sum \bigcirc + \sum_{i<j} \sum \bigcirc + \sum_{i>j} \sum \bigcirc$$

$$\sum_{j=0}^{n}\sum_{i=0}^{n} {}^{n}C_i \, {}^{n}C_j, \quad (i=j)$$

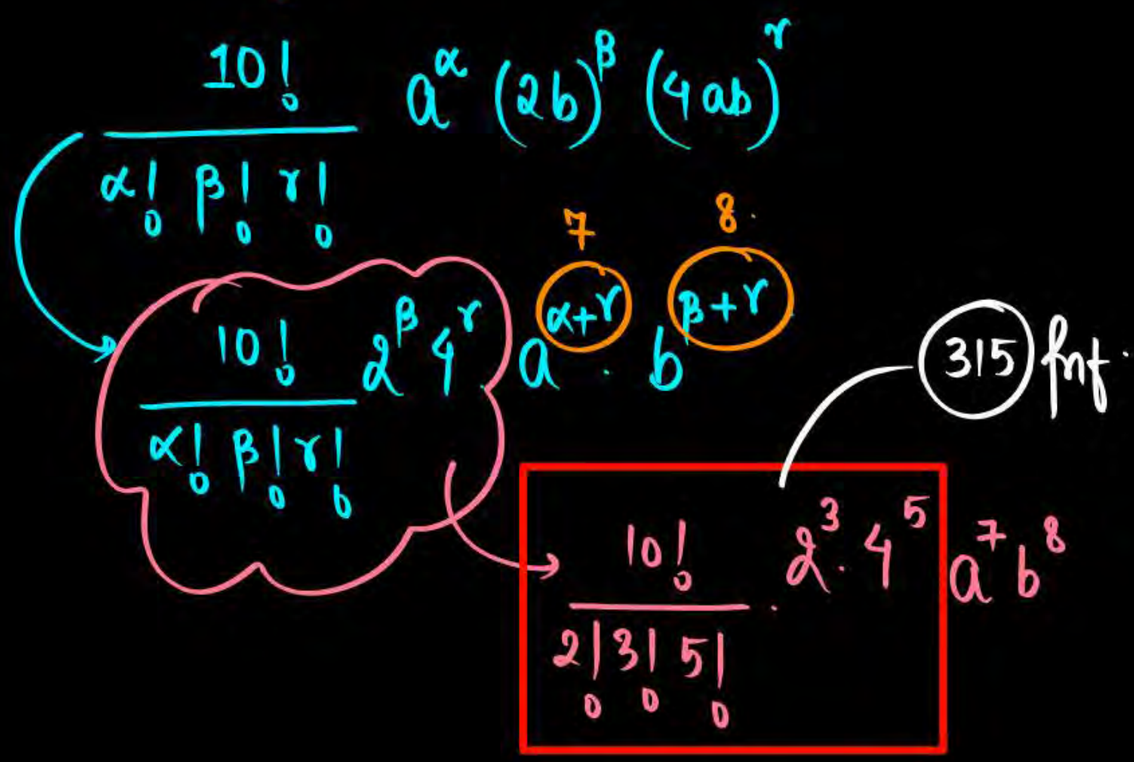
$j \to i$

$$\sum_{i=0}^{n} ({}^{n}C_i)^2 = {}^{2n}C_n$$

$$({}^{n}C_0)^2 + ({}^{n}C_1)^2 + ({}^{n}C_2)^2 + \cdots + ({}^{n}C_n)^2$$

Question [JEE Mains-2021]

If the coefficient of a^7b^8 in the expansion of $(a + 2b + 4ab)^{10}$ is $K.2^{16}$, then K is equal to_____.



$\alpha+\beta+\gamma=10$
$\alpha+\gamma=7$
$\beta+\gamma=8$

$\alpha=2$
$\beta=3$
$\gamma=5$

Question [JEE Mains-2020]

Let $(2x^2 + 3x + 4)^{10} = \sum_{r=0}^{20} a_r x^r$. Then $\frac{a_7}{a_{13}}$ is equal to ____8____.

M-I

$$\frac{10!}{\alpha!\,\beta!\,\gamma!}(2x^2)^{\alpha}(3x)^{\beta}4^{\gamma} = \frac{10!}{\alpha!\,\beta!\,\gamma!}2^{\alpha}3^{\beta}4^{\gamma}\cdot x^{2\alpha+\beta}$$

a_7 = coeff of x^7 $\Rightarrow$ $2\alpha+\beta = 7$ & $\alpha+\beta+\gamma = 10$.

a_{13} = coeff. of x^{13} $\Rightarrow$ $2\alpha+\beta = 13$ & $\alpha+\beta+\gamma = 10$.

Lengthy

14-2

Replacement

$$(2x^2+3x+4)^{10} = a_0 + a_1x^1 + a_2x^2 + \dots\dots\dots + a_{19}x^{19} + a_{20}x^{20}$$

$x \to \frac{4}{2x}$

OR

$x \to \frac{2}{x}$

$$\frac{2^{10}(4+3x+2x^2)^{10}}{x^{20}} = \frac{a_0x^{20} + a_1 2^1 x^{19} + a_2 2^2 x^{18} + \dots\dots\dots + a_{19}2^{19}x^1 + a_{20}2^{20}}{x^{20}}$$

$$(4+3x+2x^2)^{10} = \frac{a_0}{2^{10}}x^{20} + \frac{a_1 2^1}{2^{10}}x^{19} + \frac{a_2 2^2}{2^{10}}x^{18} + \dots\dots\dots + \frac{a_{19}2^{19}}{2^{10}}x^1 + \frac{a_{20}2^{20}}{2^{10}}$$

$$a_0 = \frac{a_{20}2^{20}}{2^{10}} \qquad a_{19} = \frac{a_1 2^1}{2^{10}} \qquad a_{20} = \frac{a_0 2^0}{2^{10}}$$

$$\Longrightarrow a_{13} = \frac{a_7 \cdot 2^7}{2^{10}}$$

$$8 = 2^3 = \frac{2^{10}}{2^7} = \frac{a_7}{a_{13}}$$

$$a_7 = \frac{a_{13}2^{13}}{2^{10}}$$

$$\frac{a_7}{a_{13}} = 2^3 = 8$$

APPLICATION OF BINOMIAL THEOREM

(1) Finding Remainder :-

$$7 \overline{)13}\ (2,\ 14,\ -1)$$

$$7 \overline{)13}\ (1,\ 7,\ 6) \checkmark$$

$f(7) = -1 + 7 = 6$

$13 \times 23 \times 65 \times 101$

1 2 2 2

3

$\frac{8}{3}$

Remainder $\Rightarrow 2$ ✓

100% Ques^{n} आएगा

$R=2 \Leftarrow \frac{8}{3} \equiv 13 \times 23 \times 65 \times 101$ (exponents: $-2, 2, -1, 2$; base 3)

$R=\boxed{-1} \Leftarrow \frac{-16}{3} \equiv 13 \times 23 \times 65 \times 101$ (exponents: $-2, 2, 2, 2$; base 3)

$+3 \rightarrow \boxed{2}$ ✓

$\frac{(13)^{76}}{3} \equiv \boxed{R=1}$

$13 \times 13 \times 13 \times \ldots \times 13$ (exponents: $1, 1, 1, \ldots, 1$; base 3)

$$\frac{(14)^{76}}{3} \Rightarrow R = 1$$

Sumit

$$\frac{\overbrace{14}^{-1} \times \overbrace{14}^{-1} \times \ldots \overbrace{14}^{-1}}{3}$$

$$\frac{(-1)^{76}}{3} \Rightarrow R = 1$$

Santosh

$$\frac{\overbrace{14}^{2} \times \overbrace{14}^{2} \times \ldots \overbrace{14}^{2}}{3} \Rightarrow \frac{2^{76}}{3} = \frac{\overbrace{2}^{-1} \times \overbrace{2}^{-1} \times \ldots \times \overbrace{2}^{-1}}{3}$$

$R = 1$

RAPID FIRE ROUND

(i) Find Remainder of $2^{136} \div 7$

$$\frac{2^{136}}{7} = \frac{2^1 \cdot (8)^{45}}{7} = \frac{2 \cdot 8 \cdot 8 \cdot 8 \cdots 8}{7}$$

$R = 2$

(ii) Find Fractional part of $\frac{5^{99}}{13}$

(iii) $7^{2022} + 3^{2022}$ divided by 5.

(iv) [JEE MAINS-2022] Remainder when $(11)^{1011} + (1011)^{11}$ divided by 9.

(i)

$$2^{136} = 2^1 \cdot (2^{135}) = 2^1 \cdot (2^3)^{45} = 2^1 \cdot (7+1)^{45}$$

$$\frac{2^{136}}{7} = {}^{45}C_0\, 7^{45} \cdot 2 + {}^{45}C_1\, 7^{44} \cdot 2 + \cdots + {}^{45}C_{44}\, 7^1 \cdot 2 + {}^{45}C_{45}\, 1 \times 2$$

$$\frac{2^{136}}{7} = \text{I} + \frac{2}{7}$$

$R = 2$

RAPID FIRE ROUND

(i) Find Remainder of $2^{136} \div 7$

(ii) Find Fractional part of $\frac{5^{99}}{13}$

$$\frac{5^1 \cdot 5^{98}}{13} = \frac{5^1 \cdot 25^{49}}{13} = \frac{\overbrace{5^1}^{5} \cdot \overbrace{25}^{-1} \cdot \overbrace{25}^{-1} \cdots \overbrace{25}^{-1}}{13} \equiv \frac{-5}{13}$$

(49)

$R = -5 + 13$

$= 8$

(iii) $7^{2022} + 3^{2022}$ divided by 5.

(iv) [JEE MAINS-2022] Remainder when $(11)^{1011} + (1011)^{11}$ divided by 9.

(ii') $\left\{\frac{5^{99}}{13}\right\} = \frac{\text{Remainder}}{13} = \frac{8}{13}$.

$$y \overline{)\, x \,} \; Q \;\; R \quad \Big\} \rightsquigarrow x = yQ + R$$

$$13 \overline{)\, 5^{99} \,} \; Q \;\; 8 \quad \Big\} \rightsquigarrow \left\{\frac{5^{99}}{13}\right\} = \left\{\frac{13(Q)}{13} + \frac{8}{13}\right\} = \left\{Q + \frac{8}{13}\right\} = \left\{\frac{8}{13}\right\} = \frac{8}{13} - \left[\frac{8}{13}\right] = \frac{8}{13}$$

$\left[\frac{8}{13}\right] = 0$

$$\left\{\frac{a^n}{b}\right\} = \frac{\text{Remainder}}{b}$$

FPF

$$\{\text{Integer} + x\} = \{x\}$$

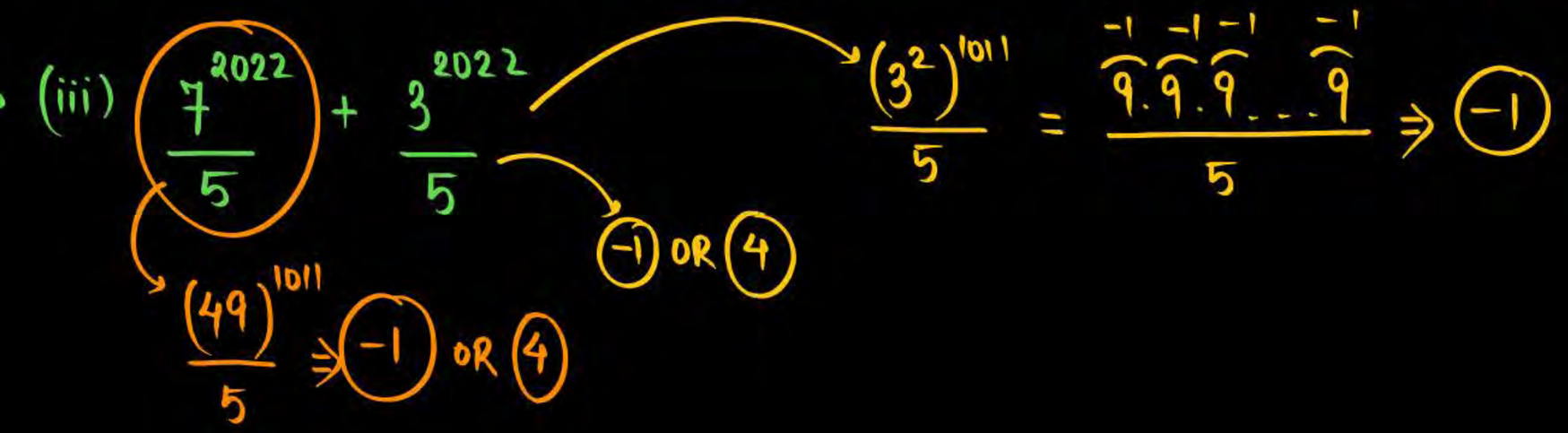
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RAPID FIRE ROUND

(i) Find Remainder of $2^{136} \div 7$

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(iv) [JEE MAINS-2022] Remainder when $(11)^{1011} + (1011)^{11}$ divided by 9.

$\frac{(11)^{1011}}{9} + \frac{(1011)^{11}}{9} \rightarrow 8 + 0 = 8$

$\frac{\overbrace{11.11.11\ldots11}^{2\ 2\ 2\ \ 2}}{9} \equiv \frac{2^{1011}}{9} = \frac{(2^3)^{337}}{9}$

$= \frac{8^{337}}{9} = \frac{\overbrace{8.8.8\ldots8}^{-1\ -1\ -1\ \ -1}}{9} \Rightarrow R = -1 + 9 = 8$

$\frac{\overbrace{1011.1011.1011\ldots1011}^{3\ \ 3\ \ 3\ \ 3}}{9} \Rightarrow \frac{3^{11}}{9}$

$R = 0$

Question [JEE Mains-2024]

The remainder when 428^{2024} is divided by 21 is

$$\frac{(428)^{2024}}{21} \equiv \frac{(8)^{2024}}{21} = \frac{(8^2)^{1012}}{21} = \frac{(64)(64)(64)\ldots(64)}{21} \Rightarrow 1$$

Question [JEE Mains-2023]

The remainder, when $19^{200} + 23^{200}$ is divided by 49, is _____.



Question [JEE Mains-2024]

Remainder when $64^{32^{32}}$ (32^{32}: k) is divided by 9 is equal to

$$\frac{\overbrace{64.64.64.\ldots.64}^{k\ \text{Time}}}{9} \Rightarrow \boxed{R = 1}$$

NCERT

Show that $6^{n+2} + 7^{2n+1}$ is divisible by 43 for all $n. \in N$

$7^{2n} \cdot 7^1 = 49^n \cdot 7^1$

$6^{n+2} + 7.(43+6)^n = 6^{n+2} + 7.\left({}^nC_0 43^n + {}^nC_1 43^{n-1} 6^1 + \ldots\ldots + {}^nC_{n-1} 43^1 \cdot 6^{n-1} + {}^nC_n 6^n\right)$

$= 6^n \cdot 6^2 + 7.\left(43(I) + 6^n\right)$

$= 6^n \cdot 36 + 7 \times 43 \times I + 7 \times 6^n$

$= 43(7I) + 6^n(36+7)$ ($36+7 = 43$)

HHP.

IMPORTANT NOTE

(a) $(x^n - y^n)$ $n \in \mathrm{N}$ is always divisible by $(x - y)$.

(b) $(x^n + y^n)$ $n \in$ odd natural number is always divisible by $(x + y)$.

$$((x+y)-(y))^n + y^n$$

$$= {}^nC_0(x+y)^n - {}^nC_1(x+y)^{n-1}y^1 + {}^nC_2(x+y)^{n-2}y^2 - \ldots\ldots + {}^nC_{n-1}(x+y)^1(-y)^{n-1} + {}^nC_n(-y)^n + y^n$$

$= (x+y) \times I$

$n \in$ odd $\Rightarrow (-1)^n y^n + y^n$

$-y^n + y^n = 0$.

$x^n - y^n$ is always div. by $(x-y)$

$25^{34} - 9^{34}$ is div. by 16

$25 - 9$

$21^{33} + 12^{33}$ is div. by 33

$(21+12)$

Question [JEE Mains-2023]

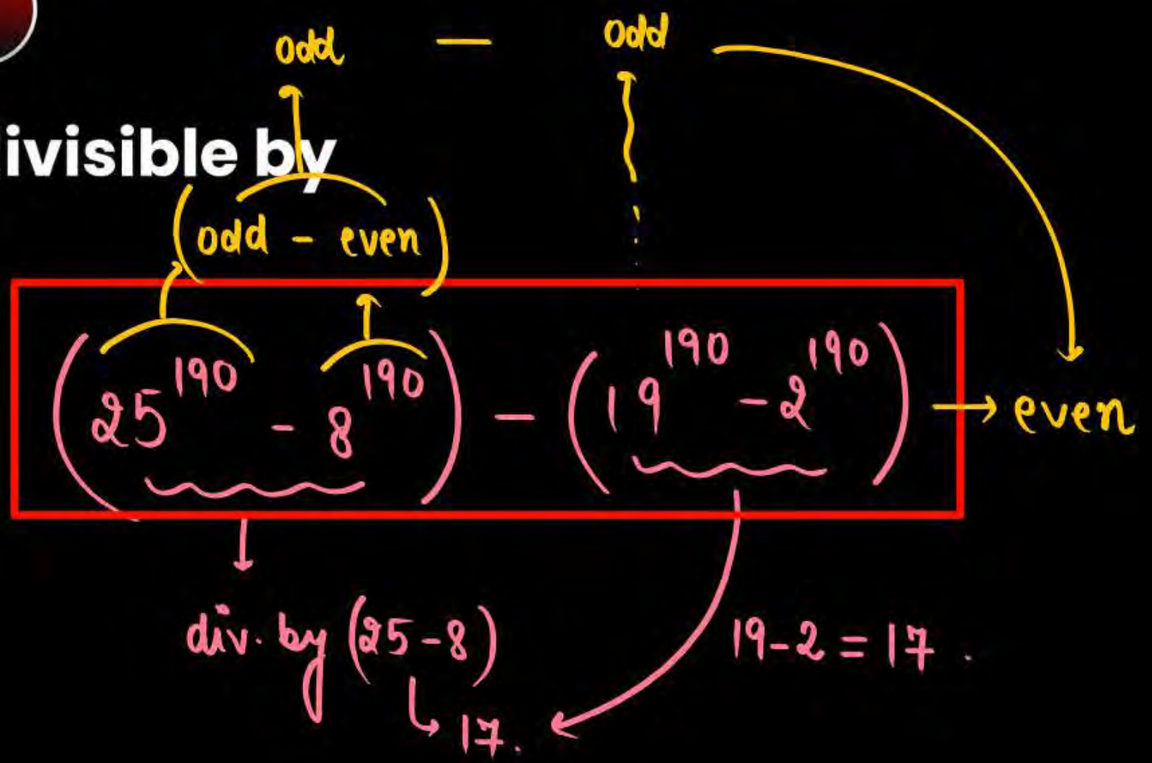
$25^{190} - 19^{190} - 8^{190} + 2^{190}$ is divisible by

A. 34 but not by 14

B. Both 14 and 34

C. Neither 14 nor 34

D. 14 but not by 34



Question [JEE Mains-2023]

Among the statements :

(S1) : $2023^{2022} - 1999^{2022}$ is divisible by 8

(S2) : $13(13)^n - 11n - 13$ is divisible by 144 for infinite many $n \in N$.



A Both (S1) and (S2) are incorrect

B Only (S1) is correct

C Only (S2) is correct

D Both (S1) and (S2) are correct

Question [JEE Mains-2023]

The remainder when $(2023)^{2023}$ is divided by 35 is _____.



FINDING LAST DIGIT(S)

Create 10

Question

Find last digit of 3^{104}.

$$3^{104} = (3^2)^{52} = (10-1)^{52} = {}^{52}C_0 10^{52} - {}^{52}C_1 10^{51} + {}^{52}C_2 10^{50} \ldots + {}^{52}C_{50} 10^2 - {}^{52}C_{51} 10^1 + {}^{52}C_{52}$$

$$= 100(I) - 520 + 1$$

$$= 100(I) - 519$$

$100(I) =$ ……00

$- \quad 519$

……81

Question [JEE Mains-2024]

Product of last two digits of 1919^{1919} is _____.



$$(1920-1)^{1919} = \text{---}$$

INTEGRAL & FRACTIONAL PART

Question

If $(\sqrt{2}+1)^6 = I + f$ where I is integral part & f is fractional part). Then find:

(i) $I = 197$.

(ii) $f^2 + 196f = 196$.

$$(\sqrt{2}+1)^6 = {}^6C_0(\sqrt{2})^6 + {}^6C_1(\sqrt{2})^5 + {}^6C_2(\sqrt{2})^4 + {}^6C_3(\sqrt{2})^3 + {}^6C_4(\sqrt{2})^2 + {}^6C_5\sqrt{2} + {}^6C_6$$

$$(\sqrt{2}-1)^6 =$$

$$(\sqrt{2}+1)^6 = {}^6C_0(\sqrt{2})^6 + {}^6C_1(\sqrt{2})^5 + {}^6C_2(\sqrt{2})^4 + {}^6C_3(\sqrt{2})^3 + {}^6C_4(\sqrt{2})^2 + {}^6C_5\sqrt{2} + {}^6C_6$$

$$(\sqrt{2}-1)^6 = {}^6C_0(\sqrt{2})^6 - {}^6C_1(\sqrt{2})^5 + {}^6C_2(\sqrt{2})^4 - {}^6C_3(\sqrt{2})^3 + {}^6C_4(\sqrt{2})^2 - {}^6C_5\sqrt{2} + {}^6C_6$$

add

$$(\sqrt{2}+1)^6 + (\sqrt{2}-1)^6 = 2\left({}^6C_0(\sqrt{2})^6 + {}^6C_2(\sqrt{2})^4 + {}^6C_4(\sqrt{2})^2 + {}^6C_6\right)$$

$$= 2\left({}^6C_0\,2^3 + {}^6C_2\,2^2 + {}^6C_4\,2 + {}^6C_6\right)$$

$$= 198$$

$(\sqrt{2}+1)^6 + (\sqrt{2}-1)^6 = 198$

$I + f + G = 198$

Int.

Integer

$0 \leq f < 1$

$0 < G < 1$

$0 < f+G < 2$.

$I + (f + G) = 198$

1

$I = 197$.

$\# \quad f+G = 1$.

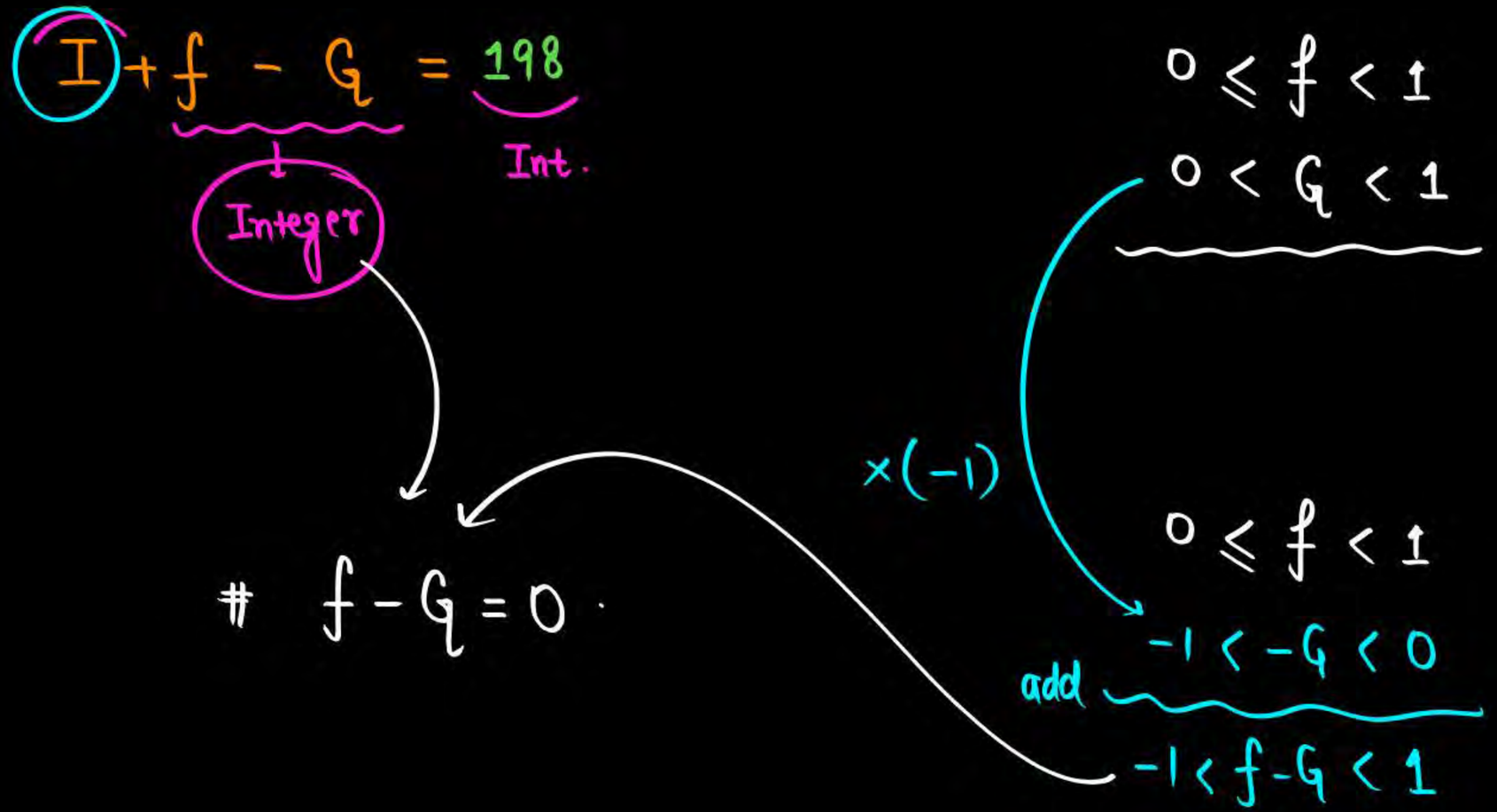
$(\sqrt{2}+1)^6 \times (\sqrt{2}-1)^6 = 1$.

$(I+f)\,G = 1$.

$(197+f)(1-f) = 1$.

$197 + f - 197f - f^2 = 1$

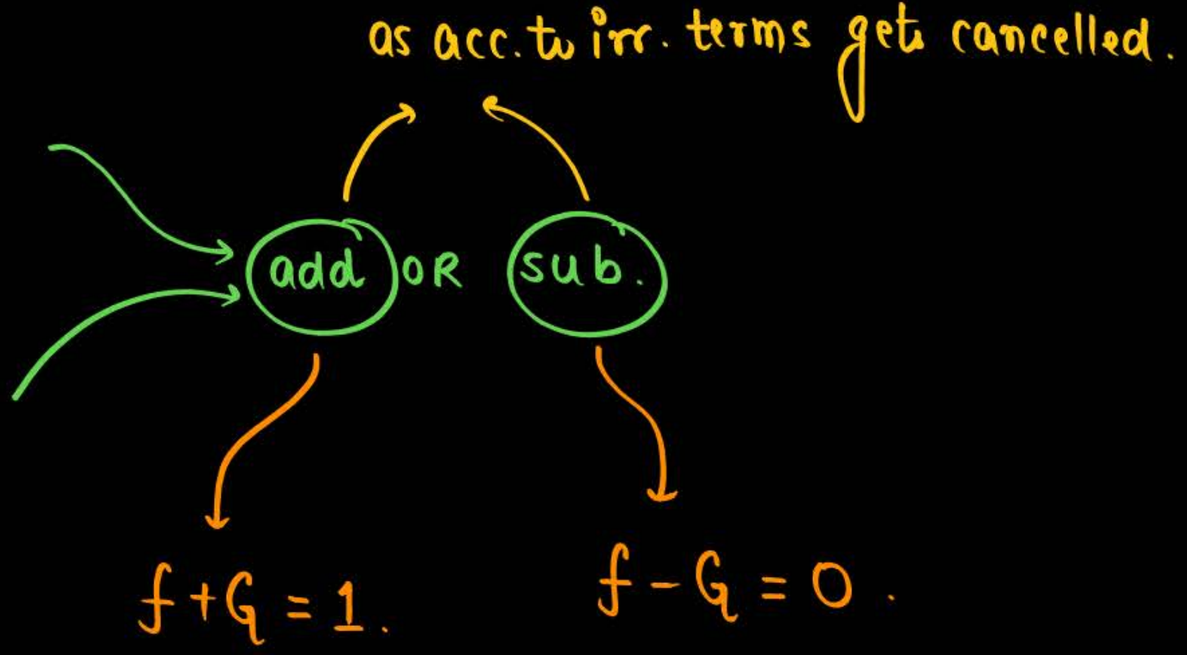
$196 = f^2 + 196f$


I + f - G = 198
Int.
Integer
0 ≤ f < 1
0 < G < 1
× (-1)
0 ≤ f < 1
-1 < -G < 0
add
-1 < f - G < 1
f - G = 0 .

S^4

$$(a+\sqrt{b})^n = I + f$$

$$(a-\sqrt{b})^n = G$$



final step: multiplication $\boxed{(a+\sqrt{b})^n (a-\sqrt{b})^n}$

Question

If $(5+2\sqrt{6})^n = I + f$, where $I \in N, n \in N$ and $0 \leq f < 1$, then prove that $I = \frac{1}{1-f} - f$.

$$(5+2\sqrt{6})^n = {}^nC_0\,5^n + {}^nC_1\,5^{n-1}(2\sqrt{6})^1 + {}^nC_2\,5^{n-2}(2\sqrt{6})^2 + \dots\dots$$

$$(5-2\sqrt{6})^n = {}^nC_0\,5^n - {}^nC_1\,5^{n-1}(2\sqrt{6}) + {}^nC_2\,5^{n-2}(2\sqrt{6})^2 - \dots\dots$$

add

$$(5+2\sqrt{6})^n + (5-2\sqrt{6})^n = 2\left({}^nC_0\,5^n + {}^nC_2\,5^{n-2}\,2^2\cdot 6^1 + \dots\dots\right)$$

$$I + f + G = I_1$$

I_1

$$\# \quad f + G = 1.$$

$$(5+2\sqrt{6})^n \cdot (5-2\sqrt{6})^n = 1$$

$$(I+f)\,g = 1$$

$$(I+f)(1-f) = 1$$

$$I = \frac{1}{1-f} - f$$

Hence.

Question

If $(\sqrt{17}+4)^7 = I + f$, then show $I = \frac{1}{f} - f$.



DO IT BY YOURSELF (DIBY)

DIBY-54

If $^{(^{30}C_1)^2} + 2^{(^{30}C_1)^2} + 3^{(^{30}C_3)^2} + \cdots .. + 30^{(^{30}C_1)^2} = \frac{\alpha \cdot 60!}{(30!)^2}$, then α is equal to

[JEE Mains-2023]

(A) 30 (B) 60 (C) 15 (D) 10

[Ans. : (C)]

DIBY-55

If a_r is the coefficient of x^{10-r} in the Binomial expansion of $(1+x)^{10}$, then $\sum_{r=1}^{10} r^3 \left(\frac{a_r}{a_{r-1}}\right)^2$ is equal to

[JEE Mains-2023]

(A) 4895 (B) 1210 (C) 5445 (D) 3025

[Ans. : (B)]

DIBY-56

If $1 + \left(2 + {}^{49}C_1 + {}^{49}C_2 + \cdots + {}^{49}C_{49}\right)\left({}^{50}C_2 + {}^{50}C_4 + \cdots {}^{50}C_{50}\right)$ is equal to $2^n \cdot m$, where m is odd, then $n + m$ is equal to ___

[JEE Mains-2022]

[Ans. : 99]

DIBY-57

If $n \geq 2$ is a positive integer, then the sum of the series ${}^{n+1}C_2 + 2({}^2C_2 + {}^3C_2 + {}^4C_2 + \cdots + {}^nC_2)$ is:

[JEE Mains-2021]

(A) $\frac{n(n+1)^2(n+2)}{12}$ (B) $\frac{n(n+1)(2n+1)}{6}$ (C) $\frac{n(n-1)(2n+1)}{6}$ (D) $\frac{n(2n+1)(3n+1)}{6}$

[Ans. : (B)]

DIBY-58

Let $\binom{n}{k}$ denote ${}^nC_k$ and $\begin{bmatrix} n \\ k \end{bmatrix} = \begin{cases} \binom{n}{k}, & \text{if } 0 \leq k \leq n \\ 0, & \text{otherwise} \end{cases}$

If $A_k = \sum_{i=0}^{9} \binom{9}{i} \begin{bmatrix} 12 \\ 12-k+i \end{bmatrix} + \sum_{i=0}^{8} \binom{8}{i} \begin{bmatrix} 13 \\ 13-k+i \end{bmatrix}$ and $A_4 - A_3 = 190p$, then p is equal to ___

[JEE Mains-2021]

[Ans. : 49]

DIBY-59

If ${}^1P_1 + 2.\,{}^2P_2 + 3.\,{}^3P_3 + \cdots .. + 15.\,{}^{15}P_{15} = {}^qP_r - s, 0 \leq s \leq 1$, then ${}^{q+s}C_{r-s}$ is equal to

[JEE Mains-2021]

[Ans. : 136]

DIBY-60

$2.\ {}^{20}C_0 + 5.\ {}^{20}C_1 + 8.\ {}^{20}C_2 + 11.\ {}^{20}C_3 + \cdots + 62.\ {}^{20}C_{20}$ is equal to [JEE Mains-2019]

(A) 2^{23} (B) 2^{26} (C) 2^{24} (D) 2^{25}

[Ans. : (D)]

DIBY-61

Let $S_n = 1 + q + q^2 + \cdots + q^n$ and $T_n = 1 + \left(\frac{q+1}{2}\right) + \left(\frac{q+1}{2}\right)^2 + \cdots + \left(\frac{q+1}{2}\right)^n$ where q is a real number and $q \neq 1$. If ${}^{101}C_1 + {}^{101}C_2 . S_1 + \ldots + {}^{101}C_{101} \cdot S_{100} = \alpha T_{100}$ then α is equal to:

[JEE Mains-2019]

(A) 2^{99} (B) 202 (C) 200 (D) 2^{100}

[Ans. : (D)]

DIBY-62

Let C_r denote the binomial coefficient of x^r in the expansion of $(1 + x)^{10}$.

If $\alpha, \beta \in R$. $1.1\ C_1 + 3.2\ C_2 + 5.3\ C_3 + \ldots\ldots$ up to 10 terms $= \frac{\alpha \times 2^{11}}{2^{\beta}-1}\left(C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \ldots \text{upto 10 terms}\right)$, then the value of $\alpha + \beta$ is equal to **[JEE Mains-2022]**

[Ans. : 286]

DIBY-63

If $\frac{1}{n+1}\ {}^nC_n + \frac{1}{n}\ {}^nC_{n-1} + \ldots + \frac{1}{2}\ {}^nC_1 + {}^nC_0 = \frac{1023}{10}$, then n is equal to **[JEE Mains-2023]**

(A) 9 (B) 8 (C) 7 (D) 6

[Ans. : (A)]

DIBY-64

Prove that : $\sum_{s=1}^{n} \sum_{r=0}^{s} {}^{n}C_s \cdot {}^{s}C_r = 3^n - 1$

DIBY-65

Find the value of $\sum_{0 \le i < j \le n} \sum ({}^{n}C_i + {}^{n}C_j)$

[Ans. : $n2^n$]

DIBY-66

Find the value of $\sum_{0 \le i < j \le n} (i + j)({}^{n}C_i + {}^{n}C_j)$

[Ans. : $n^2 2^n$]

MULTINOMIAL

DIBY-67

Find the coefficient of $a^3b^4c^5$ in the expansion of $(bc+ca+ab)^6$.

[Ans. : 60]

DIBY-68

The coefficient of x^4 in the expansion of $\left(1+x+x^2\right)^{10}$ is [JEE Mains-2020]

[Ans. : 615]

DIBY-69

The sum of all rational terms in the expansion of $\left(1+2^{1/3}+3^{1/2}\right)^6$ is equal to ___

[JEE Mains-2025]

[Ans. : 612]

DIBY-70

If the constant term in the expansion of $\left(3x^3-2x^2+\frac{5}{x^5}\right)^{10}$ is $2^k.\ell$, where ℓ is an odd integer, then the value of k is equal to: [JEE Mains-2022]

(A) 6 (B) 7 (C) 8 (D) 9

[Ans. : (D)]

FINDING REMAINDER

DIBY-71

Find remainder when 2021^{2023} is divided by 7. [JEE Mains-2022]

[Ans. : 5]

DIBY-72

The remainder, when 7^{103} is divided by 17, is [JEE Mains-2023]

[Ans. : 12]

DIBY-73

Find the remainder when

(i) 7^{103} is divided by 25. [Ans. : 18]

(ii) $2345^{678} + 1234^{567}$ is divided by 4. [Ans. : 1]

DIBY-74

Find remainder when $2021^{2022} + 2022^{2021}$ is divided by 7. [JEE Mains-2022]

[Ans. : 0]

FINDING REMAINDER

DIBY-75

Fractional part of the number $4^{2022}/15$ is equal to [JEE Mains-2023]

(A) 8/15 (B) 4/15 (C) 14/15 (D) 1/15

[Ans. : (D)]

DIBY-76

Let the number $(22)^{2022} + (2022)^{22}$ leave the remainder α when divided by 3 and β when divided by 7. Then $(\alpha^2 + \beta^2)$ is equal to [JEE Mains-2023]

(A) 20 (B) 13 (C) 5 (D) 10

[Ans. : (C)]

DIBY-77

Remainder when $2^{100} + 3^{100} + 4^{100} + 5^{100}$ is divided by 7 is

[Ans. : 5]

DIBY-78

Use binomial theorem to prove that $11^n - 10n - 1$ is divisible by 100 $\forall\ n \in$ N.



Homework

Re-attempt all Class Examples

DPP & DIBY

It is never about end result

It's always about Journey

Thank You

#futureIITians